

One-Way Over-the-Air Carrier Phase Synchronization: Design of the `ota_sync` Simulator

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Abstract

This document describes the architecture of the `ota_sync` package, a sampled-IQ simulation of one-way carrier phase and frequency synchronization between two radios (master and slave) over an NVIDIA Sionna 3GPP TR 38.901 channel. The slave disciplines its numerically controlled oscillator (NCO) using only received IQ samples. The simulator models power-law oscillator noise, a shared reference clock, accumulated timing drift, correlated shadowing, analog impairments, and correction latency, and grades every estimate against a noise-free oracle rather than against the receiver’s own measurements. At the default settings the closed-loop residual is ≈ 70 mrad RMS, set by oscillator noise and control latency rather than by SNR.

1 Overview

The package provides two models, selected by `simulation.py --model`. The *ideal* model (`ota_sync/core.py`) applies Sionna complex AWGN directly to an oscillator tone and exists as a baseline. The *SDR* model (`ota_sync/sdr.py`, the default) is the subject of this document: a framed preamble is transmitted through a tapped-delay-line multipath channel and a full analog impairment chain, and the receiver must detect the packet, acquire timing, estimate carrier frequency offset (CFO) and phase from raw IQ, and steer its NCO.

Each iteration of the main loop (`run_sdr_simulation`) represents one synchronization interval T (default 50 ms at 1 MS/s): the oscillators evolve, one frame is captured and processed, an extended Kalman filter (EKF) is updated, and a quantized correction is scheduled for the slave NCO. Defaults place the carrier at 915 MHz over a stationary TDL-D (line-of-sight) channel with 100 ns delay spread at 20 dB SNR, with an initial CFO of 1500 Hz.

2 Oscillator and clock model

Master and slave oscillators are two-state random walks (`core.Oscillator`) in phase ϕ and angular frequency ω :

$$\begin{bmatrix} \phi_{k+1} \\ \omega_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \phi_k \\ \omega_k \end{bmatrix} + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(0, \text{diag}(\sigma_\phi^2, \sigma_\omega^2)). \quad (1)$$

Three further noise processes complete a power-law phase-noise model:

White FM. A per-sample phase random walk of standard deviation σ_{pn} (default 2×10^{-4} rad) is applied inside each captured frame. Its end value, plus an independent increment for the dead time between frames, is folded back into the oscillator state, so intra-frame and inter-frame phase noise form one continuous process. Over a full interval of N_T samples this contributes phase variance $\sigma_{\text{pn}}^2 N_T$ (≈ 45 mrad RMS at defaults).

Flicker FM. A $1/f$ frequency-noise component is approximated by a bank of four first-order Gauss–Markov (AR(1)) processes with log-spaced correlation times spanning the run (`_FlickerFrequencyNoise`); the sum enters the true relative frequency each interval.

White PM. Independent per-sample phase jitter (default 5 mrad) is applied at the receiver.

Carrier LO and sample clock derive from one reference crystal. Unless a fixed override is given, the fractional sample-clock error is computed every interval from the *physical* relative oscillator frequency—the state minus all accumulated NCO corrections, since digital corrections never touch the crystal. CFO, sample-clock offset, and frame-timing drift therefore co-drift, as on real SDR hardware (1500 Hz at 915 MHz $\Rightarrow$ 1.64 ppm).

3 Frame structure and radio link

`make_sync_preamble` builds a 4606-sample frame from Zadoff–Chu (ZC) sequences: a short training field (length-16 ZC repeated 16 times) for detection and coarse CFO, followed by two cyclic-prefix-protected length-2047 ZC long training fields for fine CFO and phase.

The transmit chain (`SDRRadioLink.capture`) modulates the frame onto the master carrier, applies a soft power-amplifier limiter and DAC quantization, and passes it through per-interval Sionna TDL channel taps (`cir_to_time_channel / ApplyTimeChannel`); with nonzero mobility, Sionna evolves the taps with a Jakes Doppler spectrum. A log-normal shadowing gain with AR(1) time correlation (2 dB, 10 s defaults) multiplies the taps. The thermal noise floor is fixed once at nominal channel gain, so `snr_db` is the nominal link SNR and per-frame SNR genuinely fades.

At the receiver, the clock error accumulates into frame arrival time (≈ 0.08 samples per interval at defaults). Whole-sample drift steps the arrival inside the capture window—modeling a receiver that re-centers its window on each detection—and the sub-sample residual is applied as a fractional delay in the resampler. The samples are then downconverted by the slave LO and pass through white-FM/white-PM phase noise, IQ gain/phase imbalance, AGC, DC offset, and ADC quantization.

4 Synchronizer

`SDRSynchronizer.estimate` operates on one capture:

1. *Detection and coarse timing*: a normalized autocorrelation at the short-sequence lag; the correlation angle gives a coarse CFO estimate with unambiguous range $\pm f_s/(2 \cdot 16) = \pm 31.25$ kHz.
2. *Fine timing*: cross-correlation of the coarse-corrected window against the full known preamble; a normalized score below 0.25 declares a missed detection.
3. *Fine CFO*: the phase of the correlation between the two long fields, separated by 2175 samples, refines the CFO.
4. *Phase*: after full CFO correction, matched filtering against the long ZC sequences yields the frame phase referenced to the frame center.

5 Estimation and control

An iterated EKF (`core.PhaseFrequencyEKF`) tracks the relative state $[\phi, \omega]$ with observation $\mathbf{z} = [\cos \phi, \sin \phi, \omega]^\top$, avoiding phase-wrap discontinuities; six Gauss–Newton re-linearizations per update handle large innovations, and the Joseph form keeps the covariance positive semidefinite. The first detection initializes the state directly (acquisition), since the coarse+fine CFO estimate must resolve the cycle ambiguity before wrapped-phase tracking begins.

Both filter covariances are derived from the modeled physics. The measurement covariance adds three terms per component: the AWGN Cramér–Rao bound (≈ 1.1 mrad in phase at 20 dB), the intra-frame white-FM walk averaged over the long fields (≈ 8 mrad, dominant), and the white-PM contribution. The process covariance adds the white-FM walk over one interval and the flicker innovation to the oscillator random-walk terms.

Corrections are quantized (16-bit phase, 0.01 Hz frequency) and load into the slave NCO only after a configurable processing latency L (default one interval). The controller knows its latency and forward-predicts each command through the state transition, as a real design would. Because a command computed at frame k acts at frame $k+L$, the closed loop inherits an irreducible error from frequency uncertainty propagated over the latency.

6 Ground-truth oracle and metrics

A one-way receiver observes oscillator phase *plus* channel phase; the two are not separately identifiable, so the slave necessarily absorbs a stable channel-phase bias (about -2.9rad at defaults), which the raw oscillator residual metric exposes.

Every error metric is computed against ground truth, never against the receiver’s own measurement. Each capture is also produced along an *oracle path*: same frame, timing, channel realization, shadowing, deterministic transmit hardware, and sample-clock timing, but no AWGN, phase noise, IQ imbalance, DC offset, or ADC quantization. Running the same synchronizer on the oracle capture defines the true observable OTA phase and frequency. This matters: grading the EKF against its own input had understated the phase error by a factor of ~ 100 . Reported quantities are the tracking error (truth minus EKF estimate), the closed-loop residual (the true observable phase at each capture, which already reflects every physically loaded command), and a steady-state RMS masked to intervals after the first command loads.

7 Error budget and validation

The steady-state closed-loop residual decomposes as

$$\sigma^2 \approx \underbrace{\sigma_{\text{pn}}^2 N_T}_{\text{white-FM walk per interval}} + \underbrace{(\sigma_{\hat{\omega}} L T)^2}_{\text{freq. uncertainty} \times \text{latency}} + \underbrace{\sigma_{\text{track}}^2}_{\text{measurement/tracking}}, \quad (2)$$

where $\sigma_{\hat{\omega}}$ is the EKF steady-state frequency posterior. The simulation confirms each term in isolation (100 iterations, defaults unless noted):

Configuration	Tracking RMSE	Closed-loop residual RMS	Dominant term
Full defaults	12.0 mrad	69.6 mrad	walk (≈ 45) $\oplus$ latency (≈ 38)
No RF impairments	1.1 mrad	34.3 mrad	latency only
Zero correction latency	12.3 mrad	12.3 mrad	tracking only

Detection is 100% in all cases; frequency tracking RMSE is $\approx 0.2\text{Hz}$. The practical implication is that beyond moderate SNR the achievable phase lock is set by oscillator quality, sync interval, and control latency—not preamble energy.

8 Coherent collaboration modes

Two architectures build on the sync machinery (`ota_sync/coherent.py`), matching the two ways stations can collaborate coherently.

CSI-aided joint transmission (`--csi-gain`): when a user feeds back channel estimates, any static per-station phase bias—including the one-way channel-phase bias—is absorbed by CSI, so combining gain depends only on the *differential* phase drift between refreshes. Evaluated from the one-way run’s closed-loop residual, the two-station gain at the user stays above 99.5% even with CSI refreshed only once per second.

Open-loop coherence for passive detection (`--model twoway`): with no feedback path, the inter-station channel phase must be physically removed. A reciprocal two-way exchange—forward and reverse frames traverse the same channel realization and shadowing state—cancels it: the half-difference of the two directional phase measurements observes the pure oscillator offset, making absolute carrier alignment identifiable. At defaults this yields an 82 mrad steady-state oscillator residual (versus the -2.9rad unresolvable bias of the one-way loop) and a mean two-station open-loop coherent gain $\cos^2(\phi_e/2)$ of 99.8%. Two limits are modeled honestly: the half-difference carries a global π ambiguity (resolved in hardware by a one-time combining check), and residual TX/RX chain asymmetry (`twoway_chain_asymmetry_deg`) does not cancel in a two-way exchange and must come from loopback calibration.

9 Usage

```
python -m pip install -r requirements.txt    # sionna-no-rt==2.0.1
python simulation.py --plot                  # phase acquisition and residual
python simulation.py --plot-all             # full radio diagnostics
python simulation.py --snr-db 8 --cfo-hz 10000 --tdl-model C
python simulation.py --correction-latency 0 --no-rf-impairments
python simulation.py --model twoway          # open-loop coherence (reciprocal)
python simulation.py --csi-gain              # JT gain vs. CSI refresh cadence
python -m pytest                            # 11 tests, incl. oracle consistency
```

Key switches: `--sfo-ppm` forces a fixed sample-clock offset instead of the shared-reference derivation; `--flicker-std-hz` and `--shadowing-std-db` scale the flicker and shadowing processes; `--model ideal` runs the AWGN baseline.